

## BRIDGEPORT, TX, USA

WMO: 720261

Lat: 33.175N

Lon: 97.828W

Elev: 259

StdP: 98.25

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53976

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -6.5       | -4.1 | -13.2                         | 1.2 | -2.6 | -11.0 | 1.5 | 2.1  | 10.0                  | 10.3 | 9.1 | 10.9 | 2.2                      | 0    | 0.441 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.6                   | 39.0            | 22.0 | 37.5 | 22.1 | 36.0 | 22.4 | 25.2                | 32.4 | 24.8 | 31.7 | 24.3 | 30.9 | 3.5                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 23.6                            | 19.1 | 27.0 | 23.0 | 18.3 | 26.5 | 22.6 | 17.8 | 26.6 | 78.4          | 32.6 | 76.3 | 31.6 | 74.4 | 30.9 | 27.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 8.9               | 8.0  | 7.2 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -10.5                      | 40.1 | 2.2                | 2.1 | -12.1                                              | 41.6 | -13.4      | 42.9 | -14.6      | 44.0 | -16.2      | 45.5 |  |  |
|                   |      |     | -10.8                      | 25.9 | 2.0                | 1.0 | -12.2                                              | 26.6 | -13.4      | 27.2 | -14.6      | 27.8 | -16.0      | 28.5 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.2   | 7.0   | 9.2   | 14.0  | 17.7  | 22.1  | 26.9  | 28.5  | 28.8  | 24.6  | 18.5  | 12.7  | 7.7  |
|                                                                                  | DBStd       | 8.96   | 5.46  | 6.14  | 5.53  | 4.54  | 4.03  | 2.71  | 2.73  | 3.05  | 3.35  | 4.87  | 5.21  | 5.28 |
|                                                                                  | HDD10.0     | 378    | 124   | 82    | 25    | 4     | 0     | 0     | 0     | 0     | 0     | 3     | 32    | 108  |
|                                                                                  | HDD18.3     | 1413   | 352   | 262   | 154   | 65    | 14    | 0     | 0     | 0     | 1     | 58    | 178   | 330  |
|                                                                                  | CDD10.0     | 3366   | 32    | 59    | 149   | 234   | 374   | 508   | 573   | 583   | 437   | 267   | 113   | 38   |
|                                                                                  | CDD18.3     | 1361   | 1     | 5     | 20    | 46    | 130   | 258   | 315   | 324   | 188   | 63    | 9     | 2    |
|                                                                                  | CDH23.3     | 16428  | 19    | 60    | 196   | 452   | 1319  | 3057  | 4153  | 4244  | 2087  | 719   | 104   | 18   |
|                                                                                  | CDH26.7     | 8284   | 1     | 12    | 45    | 124   | 507   | 1537  | 2368  | 2447  | 983   | 238   | 21    | 1    |
| Wind                                                                             | WSAvg       | 3.0    | 3.1   | 3.4   | 3.7   | 3.7   | 3.3   | 3.1   | 2.5   | 2.4   | 2.2   | 2.6   | 2.9   | 2.9  |
| Precipitation                                                                    | PrecAvg     | 890    | 43    | 61    | 77    | 85    | 94    | 97    | 58    | 57    | 92    | 97    | 61    | 54   |
|                                                                                  | PrecMax     | 1205   | 91    | 187   | 206   | 273   | 172   | 310   | 120   | 148   | 223   | 240   | 158   | 213  |
|                                                                                  | PrecMin     | 637    | 3     | 0     | 1     | 7     | 30    | 30    | 3     | 1     | 14    | 12    | 2     | 0    |
|                                                                                  | PrecStd     | 154    | 26    | 51    | 48    | 56    | 48    | 67    | 41    | 41    | 57    | 74    | 53    | 47   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 26.1  | 28.0  | 30.5  | 32.8  | 35.3  | 38.1  | 41.2  | 41.3  | 37.3  | 33.6  | 29.3  | 26.0 |
|                                                                                  |             | MCWB   | 13.7  | 16.2  | 17.5  | 18.6  | 22.4  | 22.3  | 21.7  | 21.5  | 21.0  | 21.0  | 18.0  | 16.0 |
|                                                                                  | 2%          | DB     | 22.5  | 25.1  | 27.7  | 30.1  | 32.9  | 36.6  | 39.0  | 39.6  | 35.0  | 31.3  | 26.2  | 22.4 |
|                                                                                  |             | MCWB   | 13.0  | 15.4  | 17.0  | 19.0  | 21.8  | 22.1  | 21.8  | 21.7  | 21.9  | 20.8  | 16.2  | 14.2 |
|                                                                                  | 5%          | DB     | 19.8  | 22.4  | 25.4  | 27.7  | 31.3  | 35.1  | 37.4  | 37.7  | 33.5  | 29.3  | 23.9  | 19.9 |
|                                                                                  |             | MCWB   | 12.8  | 14.6  | 16.6  | 18.7  | 20.9  | 22.6  | 22.0  | 22.5  | 22.0  | 20.0  | 16.3  | 13.7 |
|                                                                                  | 10%         | DB     | 17.3  | 19.9  | 23.0  | 26.0  | 29.8  | 33.5  | 35.7  | 36.1  | 32.2  | 27.5  | 21.9  | 17.4 |
|                                                                                  |             | MCWB   | 11.0  | 13.6  | 15.5  | 17.9  | 20.9  | 22.9  | 22.5  | 22.6  | 21.6  | 18.7  | 15.5  | 12.0 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 17.9  | 19.0  | 20.2  | 22.6  | 24.9  | 26.2  | 26.2  | 25.6  | 25.2  | 23.8  | 21.2  | 19.1 |
|                                                                                  |             | MCDB   | 20.6  | 23.4  | 26.1  | 27.3  | 31.9  | 32.3  | 33.2  | 33.2  | 31.6  | 29.3  | 24.8  | 22.2 |
|                                                                                  | 2%          | WB     | 15.6  | 17.8  | 19.1  | 21.4  | 23.8  | 25.0  | 25.2  | 25.1  | 24.4  | 22.7  | 19.2  | 16.9 |
|                                                                                  |             | MCDB   | 19.4  | 22.0  | 24.5  | 26.1  | 29.9  | 31.5  | 32.4  | 32.6  | 30.1  | 27.8  | 22.2  | 20.6 |
|                                                                                  | 5%          | WB     | 13.4  | 16.4  | 18.2  | 20.4  | 23.0  | 24.3  | 24.6  | 24.6  | 23.8  | 21.8  | 18.0  | 14.9 |
|                                                                                  |             | MCDB   | 18.9  | 20.5  | 23.0  | 25.2  | 28.4  | 30.8  | 31.4  | 32.0  | 29.1  | 26.9  | 21.5  | 17.8 |
|                                                                                  | 10%         | WB     | 11.2  | 14.6  | 17.3  | 19.5  | 22.2  | 23.8  | 24.0  | 24.0  | 23.1  | 20.7  | 16.7  | 12.5 |
|                                                                                  |             | MCDB   | 16.6  | 19.2  | 21.7  | 24.5  | 27.5  | 30.2  | 30.5  | 31.2  | 28.2  | 25.4  | 21.1  | 16.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 13.2  | 12.7  | 13.1  | 13.0  | 12.0  | 11.6  | 12.7  | 12.6  | 12.8  | 13.7  | 13.4  | 12.5 |
|                                                                                  |             | MCDBR  | 17.7  | 16.7  | 15.5  | 15.0  | 14.2  | 13.2  | 14.8  | 14.2  | 14.2  | 15.1  | 16.1  | 16.9 |
|                                                                                  |             | MCWBR  | 10.1  | 9.2   | 7.5   | 7.3   | 5.2   | 3.3   | 3.5   | 3.3   | 4.3   | 6.5   | 8.6   | 9.8  |
|                                                                                  | 5% WB       | MCDBR  | 15.4  | 14.4  | 12.0  | 11.4  | 11.1  | 11.3  | 11.6  | 12.0  | 11.2  | 11.8  | 11.3  | 12.8 |
|                                                                                  |             | MCWBR  | 10.3  | 9.5   | 6.6   | 6.8   | 5.4   | 4.0   | 3.7   | 3.5   | 4.0   | 5.8   | 8.2   | 9.4  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.305  | 0.315 | 0.348 | 0.382 | 0.418 | 0.431 | 0.433 | 0.427 | 0.394 | 0.353 | 0.326 | 0.310 |      |
|                                                                                  | taud        | 2.510  | 2.488 | 2.417 | 2.344 | 2.308 | 2.308 | 2.342 | 2.337 | 2.379 | 2.478 | 2.490 | 2.495 |      |
|                                                                                  | Ebn at Noon | 916    | 941   | 930   | 906   | 872   | 856   | 852   | 854   | 870   | 885   | 881   | 888   |      |
|                                                                                  | Edh at Noon | 85     | 96    | 111   | 125   | 131   | 131   | 126   | 124   | 114   | 96    | 86    | 81    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.96   | 3.52  | 4.63  | 5.61  | 6.02  | 6.74  | 6.80  | 6.23  | 5.20  | 4.07  | 3.13  | 2.58  |      |
|                                                                                  | RadStd      | 0.30   | 0.54  | 0.33  | 0.46  | 0.61  | 0.62  | 0.45  | 0.38  | 0.46  | 0.61  | 0.40  | 0.32  |      |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (44 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page